Nikhilesh Vijay Chirutala

Full-Stack Software Engineer

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SUMMARY

Full-Stack Software Engineer with 3 years of experience building and maintaining production applications across AI, fintech, CRM, and cloud contact-center platforms. Experienced in end-to-end application development, including RESTful API development with Python and FastAPI, React.js/Next.js interfaces, SQL and data modeling with PostgreSQL and MongoDB, and authentication and RBAC/ACL authorization. Built and integrated a RAG-based chatbot into a live trading platform for contextual, portfolio-aware financial insights. Focused on debugging, production reliability, API integration, and observability using Datadog and Sentry.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Backend & Frameworks: Python, FastAPI, REST APIs, API Development, Flask, Backend Automation

Frontend: React.js, Next.js, JavaScript, TypeScript, HTML5, CSS3, Tailwind CSS, Bootstrap, Jinja2, Microservices

AI & LLM: RAG, LLM Integration, Vector Search, Prompt Engineering

Databases: PostgreSQL, SQL, MongoDB, Neon DB

Cloud & DevOps: AWS, Amazon S3, Vercel, GitHub Actions, CI/CD, Linux

APIs & Integrations: Twilio, GitHub API, REST APIs, Postman, API Integration

Security & Authentication: Authentication, Authorization, RBAC, ACL, JWT

Testing & Tools: Postman, Jest, Unit Testing, Git, GitHub, Debugging

WORK EXPERIENCE

Software Engineer · **CA One Tech Cloud Inc.** Apr 2026 – Present | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform *Python, FastAPI, AWS, Amazon S3, MySQL, Datadog*

- Develop and maintain production-grade Python services for a high-volume cloud contact center platform, supporting call-recording ETL, data processing, storage workflows, and production operations.
- Migrated a legacy Perl-based call-recording workflow to a long-running Python service deployed across 40 production clusters, supporting feature flags, staged rollout, validation, rollback, and monitoring.
- Built Python integrations with a Call Intelligence microservice, implementing service APIs, eligibility checks, account and campaign controls, daily processing limits, retries, response validation, and result persistence.
- Developed validation and recovery workflows for call-recording data, handling incomplete processing, failed operations, and inconsistent storage states across production environments.
- Supported production deployment and rollout of Python services across multiple clusters, including feature-flag configuration, validation, monitoring, and troubleshooting during migration activities.
- Built AWS S3 audit and recovery workflows for recording storage, including deterministic object handling and idempotent re-upload operations for incomplete or failed storage migrations.
- Implemented production observability using Datadog, Grafana, dashboards, and Sentry for monitoring service health, tracking errors, diagnosing processing failures, and supporting production incident resolution.
- Investigated production issues across Python services, APIs, databases, and cloud storage, performing troubleshooting and implementing fixes to maintain reliable recording-processing workflows.

Software Engineer · **Mantra Technologies** Oct 2023 – Mar 2026 | Hyderabad, India

- Delivered production features and enhancements across multiple applications, translating product requirements into application modules, business logic, and complete end-to-end workflows.
- Worked across application layers to implement and integrate functionality, connecting frontend workflows, backend services, APIs, and data flows to deliver cohesive product features.
- Implemented authentication, authorization, and access-control workflows, while integrating external services and handling validation and integration-related issues.
- Supported production releases and resolved application issues through testing, debugging, root-cause analysis, deployment validation, and defect resolution.

Key Projects:

AI Bull — AI-Powered Trading & Investment Platform *React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM*

- Developed an AI-powered trading and investment platform with portfolio aggregation, enabling users to consolidate holdings from multiple Demat accounts such as Upstox, Zerodha, and Groww into a unified view.
- Built portfolio workflows for tracking positions, stocks, and mutual funds, along with detailed reports and investment-level information across aggregated accounts.

- Developed stock and mutual fund screeners with configurable filtering capabilities and detailed asset pages for analyzing individual investments and their underlying information.
- Designed and implemented a backtesting engine for evaluating trading strategies against historical market data and analyzing strategy performance.
- Implemented a Retrieval-Augmented Generation (RAG) based chatbot that combines portfolio and financial information to provide contextual insights and answer portfolio-related user queries.
- Built FastAPI and Python backend APIs and business logic supporting portfolio aggregation, investment analysis, screening, backtesting, and AI-powered portfolio workflows.

CRM.io — AI-Driven CRM System *FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, WebSockets, AWS S3, AWS Lambda, FCM*

- Developed backend modules and REST APIs for an AI-driven CRM using Python and FastAPI, implementing business logic and workflows across Contacts, Tasks, Deals, Documents, communication, and account management.
- Built the Contacts management experience with inline editing, global search across CRM modules, access to related Tasks, Deals, and Documents, and Google Sheets integration for importing contact records in bulk.
- Developed a real-time CRM dialer accessible from contact records, integrating Twilio and an internal SIP-based telephony application through WebSockets, with AI-generated call scripts provided for relevant calls.
- Implemented a CRM-wide soft-delete and Trash workflow with a 30-day retention period, replacing hard-delete behavior and using an AWS Lambda service to permanently remove records after the retention period.
- Built document management workflows using Amazon S3, including browser-to-S3 client uploads and storage of documents associated with CRM records.
- Developed a task cadence system that allows users to create sequences of tasks as a single workflow and progress through scheduled follow-ups, with Firebase Cloud Messaging (FCM) used for task notifications.
- Built account-level reporting dashboards and developed an MCP integration exposing CRM capabilities as tools, enabling the AI bot to retrieve CRM information and trigger supported operations.

PERSONAL PROJECTS

JS Studio — Online JavaScript Playground *Next.js, GitHub API, Vercel*

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

AnyScrum — Project Management Platform *Next.js, Neon DB, Clerk, Vercel*

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

EDUCATION

B.Tech — Electrical & Electronics Engineering 2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur